

TWO-STEP BAA, CALL ANNOUNCEMENT

BAA-25-R-C2CFT, WHITE PAPER CALL # 0002

Title: Mounted and Dismounted Radio Frequency Unit (MDRFU) - Unified Intelligent and Highly Adaptive Antenna Framework

Date: 14 April 2025

FEDERAL AGENCY NAME: Army Contracting Command-Aberdeen Proving Ground (ACC-APG) on behalf of the Army Futures Command (AFC), Command and Control Cross-Functional Team (C2-CFT)

BROAD AGENCY ANNOUNCEMENT NUMBER: BAA-25-R-C2CFT

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CALL ANNOUNCEMENT NUMBER: 0002

CALL SCHEDULE: Dates:TBD

Event
BAA Released on SAM.gov
Call: TBD
White Paper (WP) Submission Opens
WP Submission Closes (TBD)
Government begins notifying Vendors of WP evaluations:
- Selected for Further Consideration (Invitation to participate in the no-cost technological demonstration) - Not Selected for Further Consideration
Technological Demonstration
Government begins notifying Vendors of results of technological demonstration:
- Selected for Further Consideration (Invitation to submit proposal) - Not Selected for Further Consideration
Government begins accepting proposals from selected Vendors
Government begins notifying Vendors selected for award (if any)

**White papers received after the due date/time shall be governed by the provisions of FAR 52.215-1(c) (3).*

BAA TYPE: This is a call for WPs only and will be governed by the two-step process described in General Information of the BAA-25-R-C2CFT. **Offerors that submit a proposal without first submitting a WP will not be eligible for award.**

WP INSTRUCTIONS: Detailed instructions on format for WPs and the two-step submission process can be found in the overarching BAA.

Important

Please assess the entire solicitation carefully prior to any response. The C2-CFT BAA utilizes a two-phase process to solicit offers and select a source for award.

- **Phase One – Step One: Submission of White Papers**
 ***Step Two: Technological Demonstration (By Invitation Only)**
- **Phase Two – Submission of Proposals (By Invitation Only)**

***Please note the Government reserves the right to skip step two of phase one and go directly to Phase Two after white paper evaluations.**

WP SUBMISSION: All WPs shall be submitted via Joint Tactical Networking Center (JTNC) Joint Communications Marketplace (JCM) Technical Exchange Meeting (TEM) Industry Community, via <https://jtnc.experience.crmforce.mil/JTNC/s/limited-public>

TECHNOLOGY AREAS OF INTEREST:

The U.S. Army Program Executive Office Command Control Communications- Network (PEO C3N), in support of Network Modernization priorities, is issuing this call for WPs in association with Topic Number 0002. To be eligible for consideration and possible contract award, the WP shall be of basic research, applied research, and/or that part of development not related to the development of a specific system or hardware procurement in the following “Technology Areas of Interest” specific to this call:

A. BAA TOPIC NUMBER: N2104: Command Post Mobility/Survivability
WHITE PAPER CALL #0002

Operational Description of the Problem:

The current Corps Command Post (CP) and On-the-Move (OTM) communications system relies on multiple radio systems and antennas. Historically, antenna farms were deployed to support dispersed operations, but the inability to dismount antennas during at-the-halt (ATH) and at-the-quick-halt (ATQH) operations limits their effectiveness. This issue is especially problematic for armored platforms operating in high-vegetation concealment areas, where the lack of repositionable antennas significantly degrades communication reliability and operational flexibility.

Technical Description of the Problem:

These antennas operate across a broad frequency range from 3MHZ to 300GHz, each equipped with its own high-power amplifier (HPA). They vary in size and height and are mounted on track and wheel vehicles. However, these communication systems face significant operational challenges, particularly for vehicles with turrets. The fixed mounting of antennas restricts their ability to function effectively during turret rotation and maneuvering.

Required Capabilities:

The current Corps Command Post (CP) and On-the-Move (OTM) communication systems rely heavily on multiple radio systems and antennas. Traditionally, large "antenna farms" were deployed to support dispersed operations. However, these systems face critical limitations:

1. Inability to dismount or reposition antennas during At-the-Halt (ATH) and At-the-Quick-Halt (ATQH) scenarios.
2. Degraded communication reliability and operational flexibility in armored platforms operating in high-vegetation or concealed environments.

3. Fixed antenna placements hinder maneuverability and concealment.
4. These issues significantly reduce the effectiveness of current communication setups, particularly in dynamic or complex battlefield conditions.

The current Corps Command Post (CP) and On-the-Move (OTM) communication systems rely heavily on multiple radio systems and antennas. Traditionally, large "antenna farms" were deployed to support dispersed operations. However, these systems face critical limitations:

1. Inability to dismount or reposition antennas during At-the-Halt (ATH) and At-the-Quick-Halt (ATQH) scenarios.
2. Degraded communication reliability and operational flexibility in armored platforms operating in high-vegetation or concealed environments.
3. Fixed antenna placements hinder maneuverability and concealment.
4. These issues significantly reduce the effectiveness of current communication setups, particularly in dynamic or complex battlefield conditions.

The solution requires a low-profile, multi-band, agile, modular, and scalable antenna system with integrated HPA functionality. Key requirements include:

1. Frequency Support (Threshold): Operates across HF, VHF, UHF, L-band, S-band, C-band, and 5G bands (30 MHz to 300 GHz).
2. Variable Power Output (Threshold): Configurable HPA output ranging from 1W to 100W.
3. Mounted/Dismounted Capability (Threshold): Quick mount/dismount features for rapid deployment.
4. Waveform Adaptability (Threshold): Support for HF, SINCGARS, TSM, 2x2 and 4x4 Silvus, and 5G (Objective).
5. System Integration (Objective): Full CMOSS and Software Defined Radio (SDR) compatibility.

Key Features & Design Objectives:

1. Multi-Band Frequency Support
 - Full-spectrum operation from HF to millimeter-wave (mmWave).
 - Simultaneous multi-frequency transmission/reception.
 - Tunable matching networks for impedance optimization and signal efficiency.
2. Agile Tuning & Frequency Adaptability
 - Development of frequency-agile antenna elements.
 - Real-time dynamic reconfiguration based on mission needs.
3. Modular & Scalable HPA System
 - Interchangeable, stackable amplifier modules for system flexibility.
 - Configurable output power tailored to specific platforms and missions.
 - Software control for power management and diagnostics.
4. Advanced Antenna Construction
 - Lightweight, low-profile materials optimized for vehicle integration.
 - Low SWAP and visual signature.
 - Designs supporting rapid mount/dismount.
 - Reliable in concealed and vegetation-heavy environments.
5. Platform Integration
 - Compact design for both mobile and turreted platforms.
 - Robust mechanical and electrical interfaces resistant to movement, shock, and vibration.
 - Plug-and-play integration with CMOSS and SDR systems.

**** The vendor shall identify any current ongoing same/similar efforts associated with the proposed white paper solution/technology area. The vendor shall provide the contract/agreement number and a brief description of the effort.**

WP QUESTIONS/ANSWERS: Government will publish WP Q&As in the JCM TEM Industry Community applicable Topic Collaboration Group area. Instructions for the Q&A session will be posted to the JCM TEM Industry Community (<https://jtnc.experience.crmforce.mil/JTNC/s/limited-public>) when the call is issued.

WP PREPARATION, SUBMISSION, AND REVIEW PROCESS: Instructions for submissions of WPs, including response content and format, are covered in Section B of the baseline BAA, BAA-25-R-C2CFT. The submitted white paper shall not exceed eight (8) pages.

TECHNOLOGY READINESS LEVEL (TRL): For all the subtopic areas listed above, the maturity of the proposed technology shall be at a TRL 6 or higher (please see enclosed TRL definitions). The end state of the effort shall be a set of demonstration/proof-of-concept systems that can be tested in a field environment.

WP EVALUATION CRITERIA:

The Government will evaluate WPs to determine which have the potential to best meet the C2-CFT's needs utilizing a two-part evaluation criterion as follows:

- **Part One (1): WPs will be evaluated using “Go” or “No-Go” gate criteria listed below.**
The “Go” or “No-Go” requirements are listed below which will determine if the WP receives further consideration. If during the evaluation one of the “Go” or “No-Go” requirements is not met, that evaluation will automatically receive a “No-Go” determination and will not receive further consideration.
 - **Cost.** The Government will review the Offeror's Rough Order of Magnitude (ROM) cost estimate for compliance with the BAA. ROM must be broken out by Base and Option, to include applicable cost elements.
 - **TRL.** The Government will review the Offeror's TRL to ensure it meets requirements as detailed in the Call for WP. Offeror must meet at a minimum TRL 6 at delivery.
 - **Schedule.** The Government will review the Offeror's proposed schedule to ensure the Offeror meets the period of performance, when the Government specifies a Not-to-Exceed schedule. The *Not-to-Exceed* schedule for this effort is 24 Months (one 12-month base period, which means vendors can deliver their prototype before 12 months, and one 12-month option period). If either the base or option period proposed exceeds 12 months, the white paper will be considered “No-Go” and not receive further consideration.
 - **Compliance with all white paper submission requirements outlined in BAA-25-R-C2CFT**
- **Part Two (2): WP will be evaluated using the following factors, which are listed in descending order of importance:**
 - **Importance to Agency Programs (Relevance/Potential for Impact).** The Government will review the Offeror's description of the technology proposed on how it solves the capability/gap/problem presented, the relevance to the Highly Desired/Desired Capabilities, scalability of the solution, and assess its worth in relation to the stated “Technology Areas of Interest.” WPs will be scored as highly likely, likely, or marginally likely to address the stated operational and technical descriptions of the problem.
 - **Technical Merit/Challenge (Innovative or Revolutionary Technology (Scientific**

& Engineering Viability)). The Government will review the Offeror's technology proposed to satisfy the technical challenges and design objectives by reviewing the Offeror's enabling technology (innovative or revolutionary approach); feasibility; technical risk and mitigation plan; alternatives (other technical data/information conveyed); technology maturation (the ability to achieve technology maturity); and time to field solution. WPs will be scored as fully or partially meeting or exceeding the stated Highly Desired and Desired capabilities.

- **Schedule.** The Offeror's schedule will be evaluated based on the reasonableness of the allocation of time to meet major activities/milestones, deliverable, metrics measures of success, potential risks, and mitigation plans. WPs will be scored as having a preferable, acceptable, or questionable schedule for delivery.

TECHNOLOGICAL DEMONSTRATION EVALUATION CRITERIA:

Note: The process outlined below is subject to change.

"Technology demonstration" entails a prototype sample presented by an offeror to showcase the proposed characteristics of the offered product described in the whitepaper as well as expand on aspects of the product that cannot adequately be described by specifications, purchase descriptions, or white paper (e.g. ease of use, quality of user experience, etc.). Demonstrations of capabilities not yet ready for a practical demonstration may be provided by an in-person paper or slide show presentation. This demonstration is also an opportunity to clarify and expand upon concepts not fully explored in the white paper and to field Government questions about the proposed prototype solution.

The Contractor must present the technology demonstration during the time specified in the Invitation to Demo. Individual time slots will be allotted to the Contractors selected to attend. If the Contractor fails to present the technology demonstration on time the Government will reject the invitation to continue through the selection process

1. Unless otherwise specified in the solicitation, technology demonstration shall be:
 - (1) Presented at no expense to the Government.
 - (2) Operated and maintained by the Contractor during technology demonstration evaluation.
2. The Government will not be liable for any damages to technology demonstration during evaluation or shipping.

Rules & Expectations

The Contractor's technology demonstration will be observed during a 45-minute session. Time slots will be provided to afford Contractor's time to prepare for the demonstration at the venue. The technology demonstration evaluation will consist of the following:

Activity	Time Allotted
Setup/Tear Down:	Up to 5 min
Technology demonstration Fixes:	10 min if needed
Presentation:	45 minutes
Total Technology demonstration Time:	45-60 Minutes

The Government will allow the Contractor to prepare their equipment in a classroom prior to beginning of the event. It is recommended not to show up more than 60 minutes prior to demonstration time to prevent crowding of the classroom.

The Government will allow each Contractor up to 5 minutes to move their equipment from the classroom to the demonstration area which is approximately 30 feet from the classroom door.

The Government will allow each Contractor up to 10 minutes to fix any problems encountered between the classroom and demonstration area. After 10 minutes has elapsed, the timer will start for the 45-minute formal evaluation.

The Government will provide the Contractors a space to deliver their technology demonstration. This area will consist of a table, podium, and a projector with screen. The Contractors should expect 3-4 Government leaders presiding over the demonstration event with supporting technical evaluators seated behind. Contractors invited to attend the event will only be allowed in the demonstration area during their allocated time until the day has concluded.

Please note, the Government reserves the right to eliminate the technical demonstration and go directly to a Request for Proposal OR hold the technical demonstration virtually.

EXPECTED NUMBER OF AWARDS: The Government reserves the right to award as many (or as few) contracts as it deems appropriate. There are no set percentages of funding allocated to each investment category.

AWARD INFORMATION: The Army may elect to award a base period with an option period as follows:

- **Base Period: Design and Build a Prototype (Up to 12 Months)**

- (a) The Offeror shall deliver the prototype(s) necessary to prove their technology proposed.
 - A Safety Assessment Report (SAR) shall be delivered with the prototype.
- (b) The Offeror shall provide training after prototype delivery that will allow the Government to conduct an operational assessment of the technology proposed.
 - In support of the training/operational assessment, the Offeror shall conduct a training for Government personnel (i.e., C2-CFT, PEO C3N, and other C5ISR representatives) onsite at Aberdeen Proving Ground (APG) and provide draft training materials.

Note: Training during Phase 1 is for the program office and support organizations only. Soldier touch-point training will occur during Phase 2 as delineated below.

- Contractor shall specify exact prototypes.

Before the completion of the base period, the Government will assess the technology proposed, looking at the broad areas of technical performance and suitability for experimentation with Army units. Based on this assessment, the Government may decide to exercise the option period.

- **Option Period: Follow-On Training and Experimentation (Up to 12 Months)**

- (a) The Contractor shall provide follow-on training and experimentation for the prototype solution at a location in the Continuous United States (CONUS) with an equipped Army unit. Contractors shall provide:

- A training and experimentation session at a U.S. military installation and/or facility. Contractor shall propose the number of days required to train an Army unit on their proposed solution(s).
- User and training manuals.
- Technical support necessary to make repairs and replace parts as required to keep the proposed technology operational throughout this phase.

The Rough-Order-of-Magnitude (ROM) shall be broken out by base and option period, as the Government may only fund the base period and not the option period.

ANTICIPATED FUNDING: The Government reserves the right to select all, part, or none of the WPs received, subject to the availability of funds. All funding is subject to change due to Government discretion and availability. Potential offerors should be aware that due to unanticipated budget fluctuations funding in any or all areas may change with little or no notice.

ANTICIPATED TYPE OF CONTRACTS/INSTRUMENTS: The Government reserves the right to award the instrument best suited to the nature of research proposed.

PERIOD OF PERFORMANCE: The contract(s) will have one (1) up to 12-month base period and one up to 12-month option period. The option period is not guaranteed and is subject to funding availability.

CALL AMENDMENTS: Offerors should monitor the beta SAM site, via <https://sam.gov/>, and Joint Tactical Networking Center (JTNC) Joint Communications Marketplace (JCM) Technical Exchange Meeting (TEM) Industry Community, via <https://jtncc.experience.crmforce.mil/JTNC/s/limited-public>, for additional notices to this Call that may permit extensions of the WP submission date or otherwise modify this announcement.

APPLICABILITY OF BASELINE BAA: All requirements of BAA-25-R-C2CFT apply. For complete information regarding BAA-25-R-C2CFT, refer to the BAA announcement as posted on SAM.gov. It contains information applicable to all calls issued under the BAA and provide information on the overall program, proposal preparation and submission requirements, proposal review and evaluation criteria, award administration, agency contacts, etc. If there is conflicting information, this call for white paper and instructions contained herein takes precedent over the BAA.